



Steak - Thermo MilliProject 1

By: Logan Teeple

Appendix: [Milliproject Appendix](#)

Background

I like to cook and find the science behind it fascinating. Many times I have wondered how the thermodynamics of cooking work. Therefore, I chose to cook steak and model the temperature of the steak. I find the perfect steak a fascinating challenge, and applying my passion for science to it seems like the perfect combination. This project is an attempt to understand steak cooking and to achieve the perfect temperature.

System Description

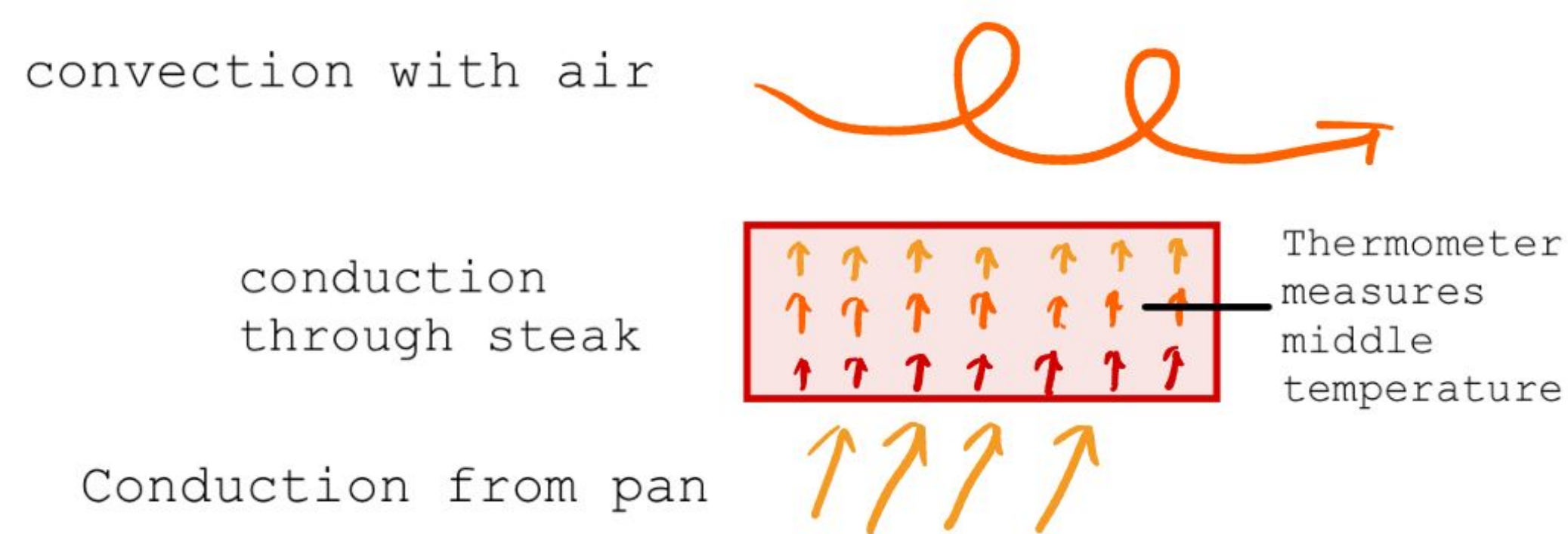


Figure 1.

System description shows the interaction between the steak with hot pan, warm air, and the conduction through itself. The thermometer was placed at about half thickness.

Model

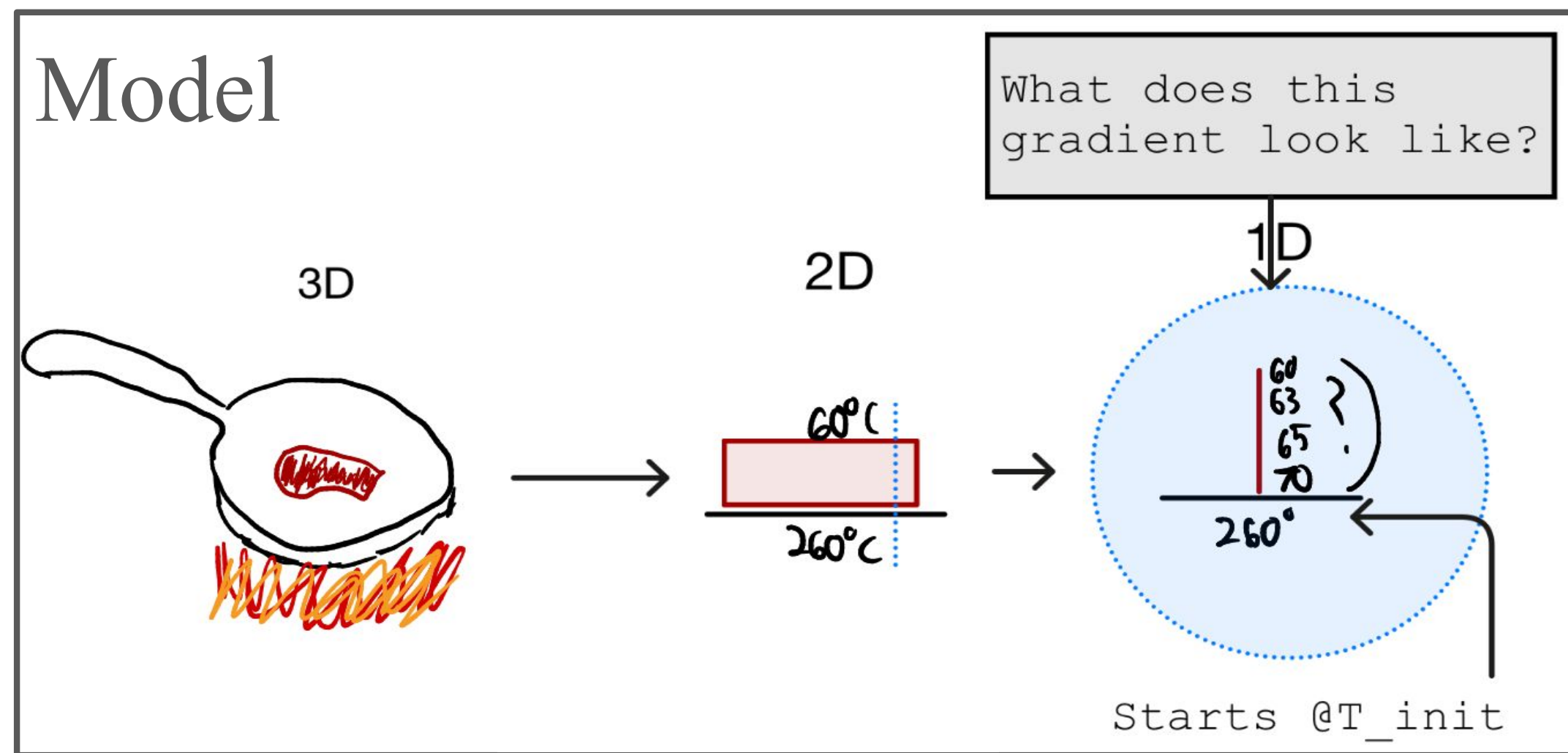


Figure 2.

The 3D reality is distilled into 1D heat conduction where the temperature gradient throughout the thickness is tracked and solved as a partial differential equation.

Model Explanation

The model uses Transient Heat transfer and the heat equation PDE because we can not assume steady state. We are trying to find out how long to cook steak for the perfect temperature, therefore steady state would return one condition which would be a burnt condition. PDE:

$$\begin{aligned} \frac{\partial u}{\partial t} &= \alpha \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < L, \quad t > 0 \\ u(x, 0) &= u_0(x), \quad 0 \leq x \leq L \\ u(0, t) &= g_0(t) \quad (\text{Dirichlet at } x=0) \\ -k \left(\frac{\partial u}{\partial x} \right)(L, t) &= h_L [u(L, t) - T_\infty, L(t)] \quad (\text{Robin at } x=L) \end{aligned}$$

For boundary conditions I used a Dirichlet for the pan as a constant temperature and a robin condition for convective boundary at the top of the steak. The pan temperature was measured as 250°C and the air was estimated at 50°C based on duration to discomfort for holding our hand above it. I simulated 40 nodes throughout the thickness and used MATLAB's pdepe PDE solver to produce a solution. The thermal values were determined from papers.

Model - Prediction

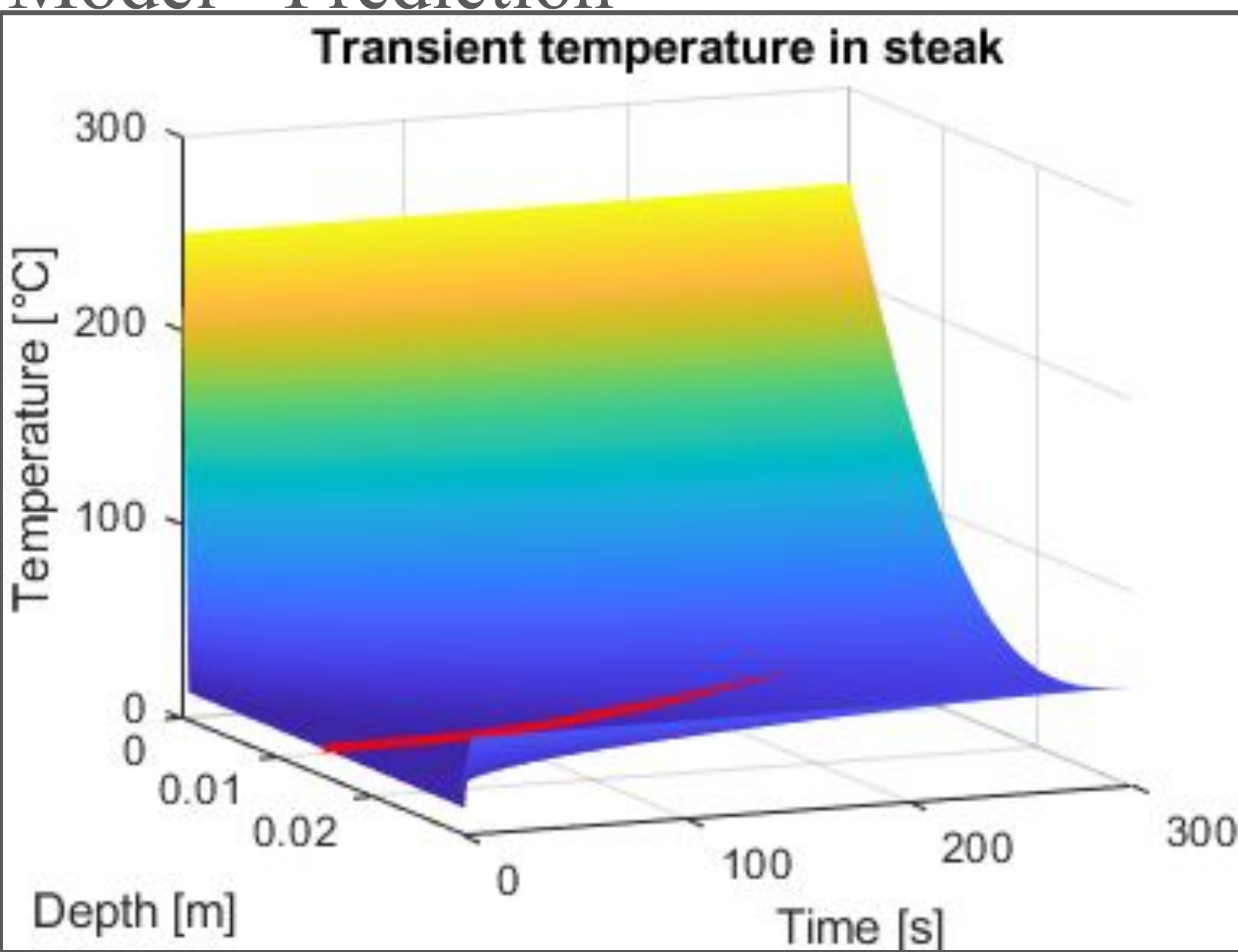


Figure 3.

Predicted scalar field of temperature over time and space. The pan boundary is the far side and the near portion is the air boundary. The line is the measured temperature halfway through.

Results

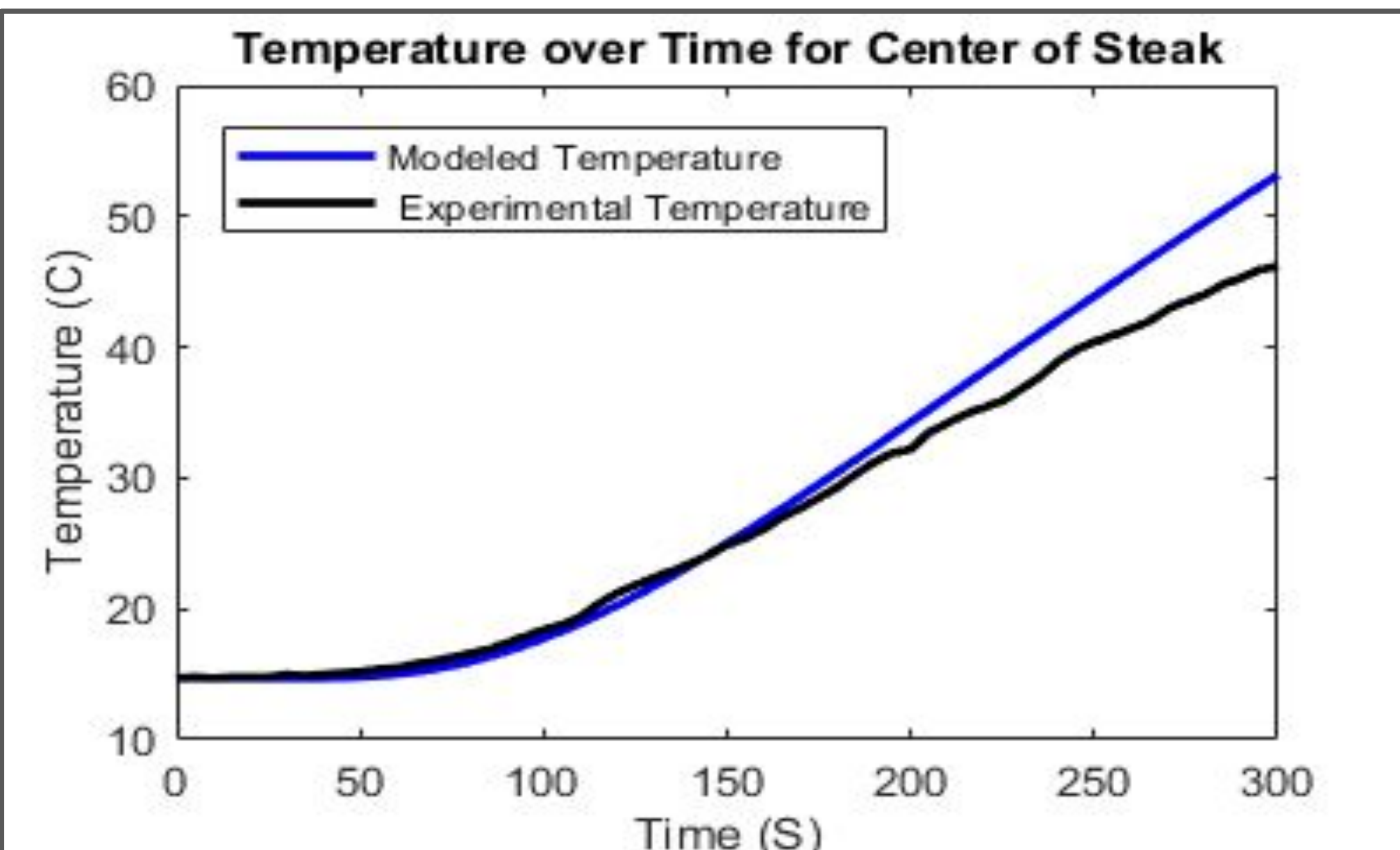


Figure 4.

Comparison of modeled and actual temperature values. The results match well until the end of the time span..

Analysis

The model produces a very rational overall scalar field (Figure 3) and the sliced comparison in Figure 4 shows that it is pretty accurate at first and then diverges at 200s. The scalar field shows that at $t = 0$, the entire profile is the initial temp, then in the next time step, the pan side becomes T_{pan} . The whole gradient warms throughout time with diffusion. I believe the diversion in Figure 4 in the later time steps is due to a change in thermal diffusivity in cooked steak. Source 1 in the appendix shows that the thermal diffusivity for steak decreases as it is cooked. My model was limited in that the thermal diffusivity was constant. Therefore, I think this is an explainable and understandable deviation. The modeled temperature is higher because it continues to use the higher thermal diffusivity value, where as the reality is the diffusion is slowing.

Conclusion and Reflection

My model and experiment gave better than expected results. I think the major false assumption was a constant thermal diffusivity over time. Despite that, my model proves useful, and with dynamic thermal diffusivity it could reduce error. This was a fascinating deep dive into heat flow and cooking!